TeachMe — Retrieve for the learner, not just the query.

TeachMe is a domain-agnostic platform for training users to operate complex, safety-critical equipment. It combines educational frameworks, agent-guided instruction, and retrieval from an expert-reviewed knowledge graph to provide grounded guidance, adaptive practice, and mastery-based progression.

A technician learning the Optomec HD2 — or a mariner learning the rules of the road — needs answers at the right level of complexity, with a guarantee that the system will never hallucinate a safety value. If a technician internalizes even one incorrect parameter during training, they may leave more dangerous than when they started.

Architecture Overview

Knowledge Graph — typed nodes and edges → **Learner Model** — proficiency + mastery gates → **Retrieval** — graph + prompt enricher → **Mentor + Narrator** — two-agent architecture → **Instructional Modes** — Socratic, Scenario, Reachback, Lab

What's Working Now

NEW: WE NOW MEASURE WHETHER THE TUTOR ACTUALLY USES ITS CORPUS (COMPANION POSTER)

01
Corpus-bound Narrator
Reports machine behavior verbatim from reviewed graph nodes; cannot generate from model weights.

→ so a learner can't internalize a hallucinated safety value; a silent corpus shows the gap, never fabricates it.

02
Typed causal graph
100+ nodes, six fault domains: Symptom → FailureMode → CorrectiveAction with SafetyHazard co-retrieved along the chain.
→ so safety context travels with the diagnosis instead of being a separate lookup a flat store can miss.

03
Dual-axis mastery gates
Declarative proficiency (Socratic probes) must clear before procedural practice; gates are transparent and auditable.

→ so an instructor can point at a score and say exactly why the system advanced or held a learner.

04
Four modes, one progression
Socratic Practice → Scenario Mode (deterministic fault injection) → Reachback Lookup → Retrieval Lab.

→ so training load is sequenced and instructors inspect what retrieval will surface before a learner sees it.

Five-Level Expertise Continuum

- **Complete Novice** — no lab background
- **Novice** — eng. tech, new domain
- **Naive** — vocabulary only
- **Intermediate** — supervised hands-on
- **Proficient** — SOPs + physics + cues

ADAPTIVE SCAFFOLDING AT EVERY LEVEL → DISTAL TRANSFER

One Engine, Three Domains

EDDIE · Aerosol Jet Printing — flagship
Optomec HD2 operator training; 100+ nodes, six fault domains, hazards co-retrieved.

Roadside Tire Change
authored as graph + corpus alone — inherits probes, scenarios, gates, consequences.

COLREG Collision Avoidance
rules of the road + interactive simulator: real kinematics, Collision Risk Index, per-rule scoring.

→ so a new domain is an authoring task, not a software project.

IF THIS WORKS

A technician reaches independent, unsupervised operation faster and with fewer escalations — because training builds far-transfer diagnostic capability, not recall.

TARGET
far-transfer performance on novel fault combinations vs. direct instruction — not yet measured on real learners; the Workflow Demo exists to de-risk this bet first.

HOW WE KEEP OURSELVES HONEST

"In safety-critical training, 'I don't know, but here's the SOP for this state' beats a confident wrong answer."

SIGNAL	WHAT IT CAN CLAIM	TRUST	CAN'T RULE OUT
Corpus-bound Narrator	Eliminates hallucinated machine behavior by construction — every observable traces to a reviewed node.	strong	states outside the authored fault domains aren't narratable (a coverage gap, made visible not hidden)
Corpus necessity audit	We test whether answers actually depend on the corpus — a calibrated screen, verified on five different base models, before any human trial.	strong	evidence is from simulation — it audits the tutor's grounding, not how well humans learn
Dual-axis learner model	Transparent, auditable tutor decisions, inspectable in the Retrieval Lab.	strong	deliberately simpler than probabilistic knowledge tracing — may miss subtler learner states
Far-transfer outcome	PBL design targets far transfer to novel fault combinations.	weak	whether the simulated-learner loop predicts real technician outcomes

LEFT OUT ON PURPOSE free-running text generation. The Narrator can only quote the reviewed corpus — so no prompt, and no jailbreak, can make it invent machine behavior.

THE QUESTIONS WE'RE CHASING

How do we effectively use AI in high-consequence environments?

Q1 Do learners trained on injected faults diagnose novel fault combinations better than those taught by direct instruction?

Q2 Does passing a mastery gate predict safe, unsupervised operation on the real equipment?

Q3 How far can the learning sciences take operational readiness in high-consequence work?

github.com/wrgr/socratic-scenarios

Mia Ndousse-Fetter William Gray-Roncal And Team